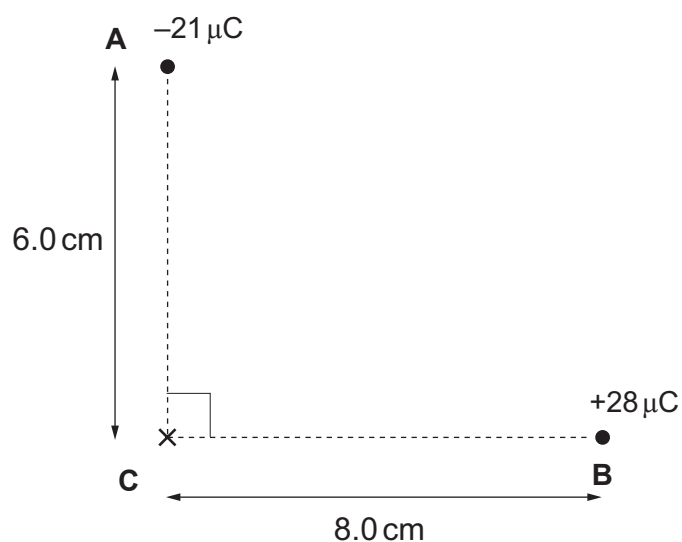


2. Two charges are placed at points **A** and **B** as shown in the diagram.



- (a) Draw **two** arrows on the diagram to represent the directions of the electric fields at **C** due to the $-21\ \mu\text{C}$ and $+28\ \mu\text{C}$ charges. [2]
- (b) Calculate the magnitude and direction of the resultant electric field at **C**. [5]

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(c) Show that the electrical potential at point **C** is zero.

[2]

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(d) A positron is initially at rest at point **C** and accelerates rapidly due to the electric field. Calculate the potential at the point where the velocity of the positron is $5.4 \times 10^7 \text{ m s}^{-1}$.

[3]

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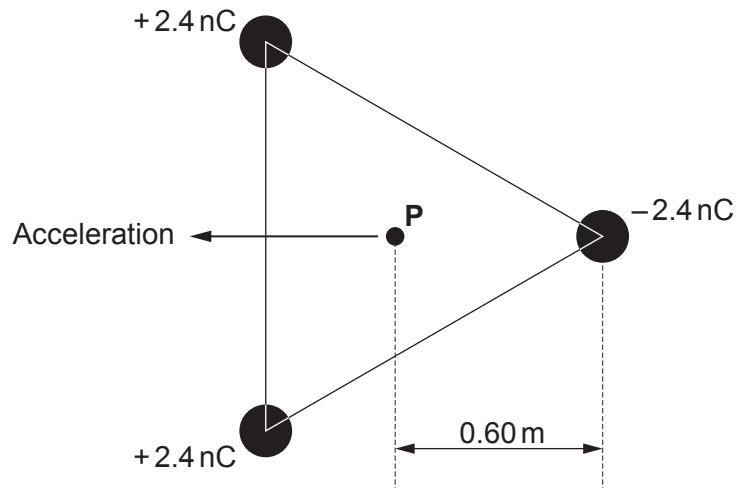
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3. Three stationary charges are located at the corners of an **equilateral** triangle and an electron is located at the centre, **P**, of the triangle as shown. The electron is exactly 0.60 m from each charge.



- (a) Draw arrows to show the 3 electric fields at **P** due to the 3 charges. Use these to explain why the electron will accelerate in the direction shown. [4]

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(b) Calculate the initial acceleration of the electron.

[4]

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(c) Show **clearly** that the initial potential energy of the electron is approximately $-6 \times 10^{-18} \text{ J}$.

[2]

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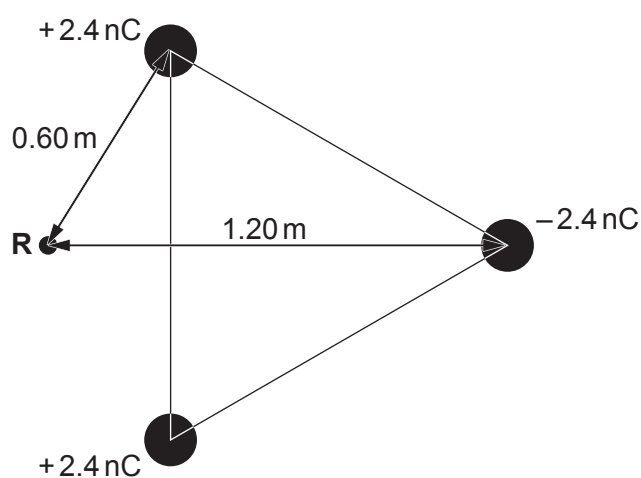
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- (d) Determine whether or not the electron (initially at rest) will reach point **R**, shown in the diagram below (both positive charges are 0.60 m from **R**). Explain your reasoning. [4]



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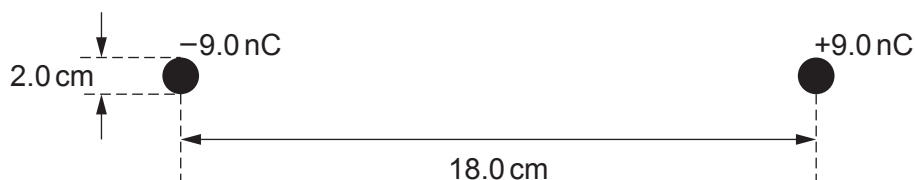
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6. Two small spheres of radius 1.0 cm carry uniform charges of +9.0 nC and -9.0 nC. Their centres are separated by a distance of 18.0 cm as shown. Treat the charges as being at the centres of the spheres.



- (a) (i) Show that the potential at the surface of the negative sphere is approximately -7500 V. [3]

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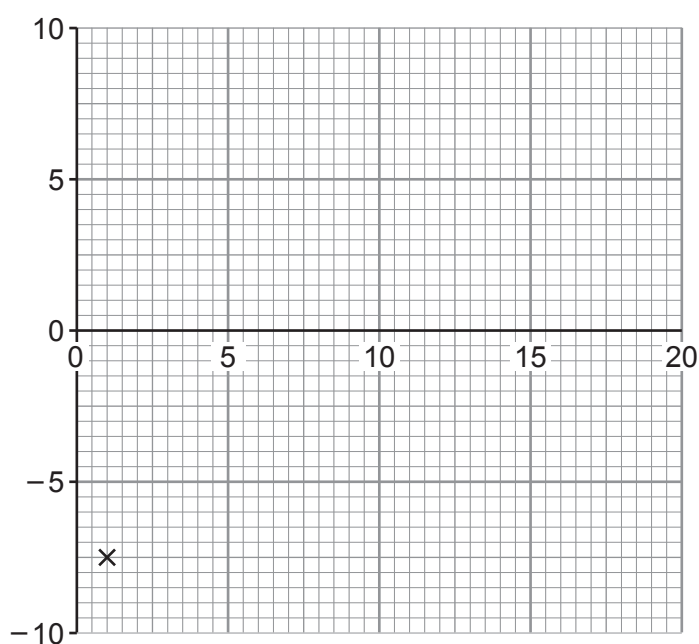
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- (ii) Sketch the graph for the electric potential between the two charges (the potential has been plotted for the surface of the negative charge). [3]

Potential / kV



Distance from
centre of
negative
charge / cm



- (b) Explain why the electric field on the right-hand surface of the negative sphere is approximately $810\,000\text{ NC}^{-1}$ and directed towards the left.

[3]

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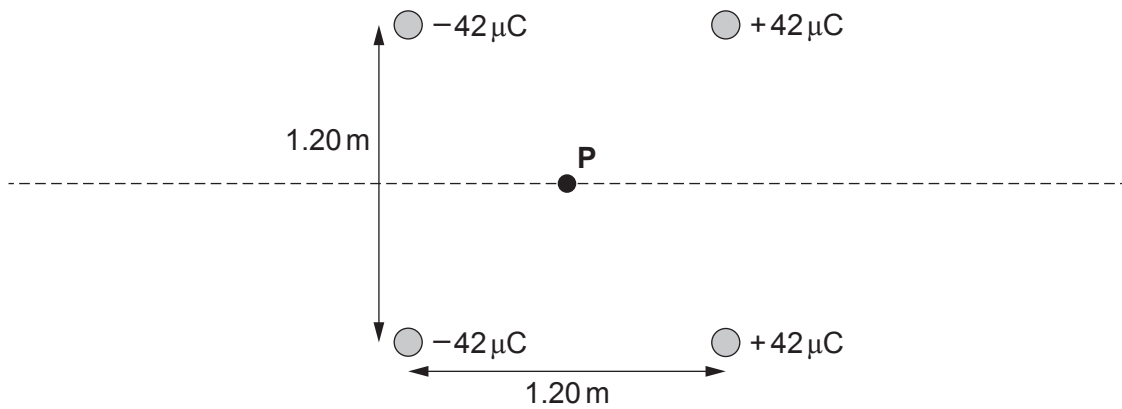
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4. Four point charges are placed at the corners of a square of length 1.20 m as shown. Point **P** is located at the centre of the square.



- (a) (i) Define electric field strength. [1]

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- (ii) Show that the magnitude of the electric field strength at **P** due to any one of the four charges is approximately 520 kNC^{-1} . [3]

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- (iii) Hence, calculate the resultant electric field strength at **P** and give its direction. [3]

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- (b) A negative point charge is held at **P** and is free to move along the dotted line shown in the diagram with no resistive forces.

(i) Define electric potential.

[1]

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(ii) Explain briefly why the potential energy of the negative charge is zero at **P**.

[2]

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(iii) Aled states “The negative charge initially accelerates to the right, then slows down but will just about reach infinity if no resistive forces act.” Discuss to what extent Aled’s points are correct.

[3]

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- (iv) A charge of -38 nC and mass of $7.7 \times 10^{-4}\text{ kg}$ is released from rest from point **P**. Calculate the speed of the -38 nC charge when it reaches point **X** shown below.

[5]

